Encyclopedia entry

Process Tracing

Sasha Kovacevic

January 2026

© 2026 Sasha Kovacevic. All rights reserved.
This copy is shared for reading only. It may not be reproduced, redistributed, modified,
or used for commercial purposes or machine-learning training without the author's written permission.
Brief quotations with full citation are permitted.
Cite as: Kovacevic, S. (2026). Process Tracing. Encyclopedia entry.

© 2026 Sasha Kovacevic. All rights reserved. Reading copy: no reproduction, redistribution or commercial use without written permission.

Process Tracing

Originating in the 1970s, process tracing is a theory-based research methodology used to shed light on the causal mechanisms underlying decisions. It does this by examining how information is searched and processed between the presentation of a stimulus and participants' final responses. In cognitive and behavioral decision research, this direct insight into how attention unfolds over time allows researchers to see which attributes people look at, how often they return to them, and how long they spend inspecting each one (Schulte-Mecklenbeck et al., 2011).

The most widely used tool in modern process tracing research is MouselabWEB. In this computer interface, task relevant information, such as outcomes or probabilities, is hidden behind boxes and only revealed when a participant moves the cursor over a box, disappearing again when the mouse is moved away. While seemingly trivial, this "mouse-over" paradigm allows precise measurement of inspection duration, frequency of access, and the sequence of information search (Willemsen & Johnson, 2011). Unlike choice data alone, these time-stamped interactions yield granular behavioral traces of pre-decisional attention.

Attentional Indices

These precise measurements allow researchers to extrapolate attentional indices. Common metrics, such as frequency, duration, and relative attention, summarize how attention is allocated. Frequency and duration are defined as how often a piece of information is accessed and how long it is inspected, respectively. Relative attention measures how much attention is given to one type of information compared to another. For example, the attentionLA index expresses the ratio of time spent on loss outcomes relative to gains. In other words, it captures bias or asymmetry in attention, not just overall effort (Pachur et al., 2018). Notably, these indices remain highly consistent over time within individuals. This indicates that attention patterns reflect enduring cognitive tendencies rather than momentary fluctuations (Glöckner & Pachur, 2012).

Processing Tracing and Cognitive Models

Acting as a bridge between mathematical models, process tracing helps connect formal decision models, such as CPT, to the underlying cognitive processes. This contribution allows for CPT parameters to be treated as potential indicators of psychological mechanisms rather than purely descriptive curve-fitting tools by demonstrating systematic associations between model parameters and attentional regularities. For example, individuals who were more loss-averse also devote relatively more attention to losses, while those with lower probability sensitivity spend less time inspecting probabilities (Pachur et al., 2018).

Significance

In short, process tracing is valued because it reveals otherwise hidden aspects of cognitive processing. By linking formal decision models to patterns of information search, it shows how stable cognitive capacities are expressed in ongoing behavior (van Rooij & Baggio, 2021).

© 2026 Sasha Kovacevic. All rights reserved. Reading copy: no reproduction, redistribution or commercial use without written permission.

References

Glöckner, A., & Pachur, T. (2012). Cognitive models of risky choice: Parameter stability and process implications. *Journal of Behavioral Decision Making, 25*(3), 220–233.

Pachur, T., Schulte-Mecklenbeck, M., Murphy, R. O., & Hertwig, R. (2018). Prospect theory reflects selective allocation of attention. *Journal of Experimental Psychology: General, 147*(2), 147–169.

Schulte-Mecklenbeck, M., Kühberger, A., & Ranyard, R. (2011). *A handbook of process tracing methods for decision research.* Psychology Press.

van Rooij, I., & Baggio, G. (2021). Theory before the test: How to build high-verisimilitude explanatory theories in psychological science. *Perspectives on Psychological Science, 16*(4), 682–697.

Willemsen, M. C., & Johnson, E. J. (2011). Visiting the decision factory: Observing cognition with MouselabWEB and other information acquisition methods. In M. Schulte-Mecklenbeck, A. Kühberger, & R. Ranyard (Eds.), *A handbook of process tracing methods for decision research* (pp. 21–41). Psychology Press.

© 2026 Sasha Kovacevic. All rights reserved. Reading copy: no reproduction, redistribution or commercial use without written permission.